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Right on time: evaluating accuracy and timing of burn-severity assessments to support post-fire management

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Abstract

Effective wildfire impact assessment and post-fire management rely on burn-severity maps that accurately characterize fire effects on vegetation and soils. These maps inform time-sensitive activities such as post-fire flood and debris-flow modeling, risk mitigation for homes and communities, and ecosystem recovery planning, but the satellite observations used to generate them are often obstructed or delayed by smoke, clouds, and snow. Across 9,129 large fires (≥ 404 ha) in Canada and the United States from 2016–2024, we quantified post-fire data obstruction and evaluated observational latency – defined as the time required to acquire the first usable scene – for four freely available moderate-resolution remote sensing datasets: Landsat, Sentinel-1, Sentinel-2, and the Harmonized Landsat Sentinel-2 (HLS) product. For multispectral datasets (Landsat, Sentinel-2, and HLS), cloud cover was the primary source of data obstruction, particularly in mesic and coastal regions, whereas Sentinel-1 (synthetic aperture radar) can penetrate clouds. Consequently, Sentinel-1 had the shortest observational latency (median = 4 days), followed by HLS (7 days), Sentinel-2 (10 days), and Landsat (17 days). Likewise, one-month composite imagery from Sentinel-1 (89.6%) and HLS (82.8%) achieved near-complete coverage for most fires, substantially better than Sentinel-2 (70.5%) and Landsat (56.1%). Despite its widespread use in operational burn-severity mapping, Landsat provided the slowest and least complete early post-fire coverage. For 77 fires in forested ecosystems, we also evaluated relationships between early post-fire imagery and field-derived burn-severity measurements collected 0–3 years after fire. All multispectral datasets corresponded well with field data and reliably identified areas of near-total canopy loss using imagery acquired within two months after fire ($> 80\%$ balanced accuracy), performing nearly as well as extended assessments using imagery from a full post-fire growing season. Sentinel-1 produced less accurate burn-severity estimates but may remain valuable where persistent cloud cover limits observations, including eastern Canada and the Pacific Northwest. Overall, HLS provided the best combination of latency, spatial coverage, and burn-severity mapping performance, making it a strong candidate for operational use and ecological monitoring across diverse North American ecosystems.

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Keywords Emergency burn-severity assessment, Initial burn-severity assessment, Post-fire remote sensing, Harmonized landsat sentinel-2, C-band synthetic aperture radar, Remotely sensed data obstruction

Introduction

Wildfire activity is shifting in response to anthropogenic climate change and altered land use practices (Jones et al. 2022; Potapov et al. 2025; Ellis et al. 2025), with widespread impacts on communities and ecosystems across North America (Coop et al. 2020; Tang et al. 2024; Carlson et al. 2025; Fortin et al. 2026). Many of the most significant socioeconomic consequences of wildfire happen after the flames have subsided (Barrett 2018; Bayham et al. 2022). Severe wildfires can alter vegetation and soil properties, thereby increasing runoff, degrading water quality (Rhoades et al. 2019; Williams et al. 2022), and accelerating erosion and sedimentation near human infrastructure (Rengers et al. 2020; Girona-García et al. 2021). Similarly, the value of timber resources often declines rapidly after fire (Prestemon et al. 2006). As a result, time-sensitive activities such as flood and debris-flow modeling, hillslope and channel stabilization, revegetation, and salvage logging are implemented immediately following fire (Danneynrolles et al. 2024; Zupan et al. 2024; Jones et al. 2025). Burn-severity maps, which quantify fire impacts on vegetation and organic matter (Keeley 2009), are critical for guiding these efforts (Robichaud et al. 2009; Stevens et al. 2021; Larson et al. 2022; Fernandez Sirgo et al. 2026). Consequently, the value of these maps depends on both rapid delivery and accurate characterization of fire effects.

Early burn-severity assessments are typically conducted within one to three months of fire and rely on moderate-resolution (i.e., 10–30 m) remote sensing data to detect fire-induced changes in vegetation and soils (Hudak et al. 2004; Miller and Quayle 2015). These methods are the

basis for several operational programs in Canada and the United States (US), including the Burned Area Reflectance Classification (BARC) and Rapid Assessment of Vegetation Condition after Wildfire (RAVG) (Key and Benson 2006). Analysts supporting these programs often use a ‘paired-scene’ approach, selecting individual pre- and post-fire images that minimize phenological differences and accentuate fire effects (M. Nigrelli, personal communication). Alternatively, an ‘image-composite’ approach can be used to automatically combine multiple images acquired within defined pre- and post-fire periods (Parks et al. 2018; Whitman et al. 2020; Vanderhoof et al. 2025). Operational mapping has historically relied on imagery from the Landsat program because of its long, consistent, and publicly available record (Wulder et al. 2022), which is well suited to burn-severity assessment (Hudak et al. 2007). Indeed, Landsat data have supported analyses of multi-decadal burn-severity trends across a range of North American ecosystems (Picotte et al. 2020; Guindon et al. 2021; Parks et al. 2025). However, its infrequent revisit interval and other sensor limitations can delay the acquisition of usable post-fire imagery, slowing map production (Orland et al. 2025). Alternative data sources, including Sentinel-1 (C-band synthetic aperture radar; SAR), Sentinel-2 (multispectral), and the Harmonized Landsat Sentinel-2 (HLS) dataset (Li and Roy 2017; Claverie et al. 2018), are being increasingly used for burn-severity mapping (Lasaponara and Tucci 2019; Alarcon-Aguirre et al. 2022; Howe et al. 2022; Park et al. 2024). Assessing how these different data sources balance timeliness and accuracy is important for improving operational post-fire mapping programs (Table 1).

Table 1 A summary of moderate-resolution, remotely sensed data sources that can support early burn-severity assessment. Revisit times give the typical number of days between acquisitions at the equator based on operational satellites in each program (as of July 2026). Delivery time gives the typical amount of time between data acquisition and public availability of surface reflectance (HLS, Landsat, Sentinel-2) or ground range detected (Sentinel-1) data. HLS: Harmonized Landsat Sentinel-2

Dataset	Wavelengths covered	Spatial resolution	Revisit time	Delivery time	Unique characteristics
HLS	13 bands; 0.43–12.51 μm ^a	30 m	2 days	3 days ^b	High-frequency acquisition
Landsat OLI	11 bands; 0.43–12.51 μm	15–60 m	8 days	4–11 days (Landsat 8); 3 days (Landsat 9)	Long temporal record when paired with TM/ETM+
Sentinel-1	1 band; 5.55 × 10 ⁴ μm	10 m	6 days	< 1 day	Can penetrate clouds and smoke
Sentinel-2	13 bands; 0.44–21.90 μm	10–60 m	5 days	< 1 day	Improved spectral, spatial, and temporal resolution relative to Landsat

^a The number of bands and wavelength ranges differ between Landsat (HLSL30) and Sentinel-2 (HLS530) components of HLS. Here, we report the maximum of both
^b Delivery time also differs between Landsat and Sentinel components of HLS, with Sentinel-2 and Landsat 9 data being processed more quickly than Landsat 8. The HLS low-latency product is forthcoming and has a target delivery time of < 6 h

The utility of these remotely sensed datasets depends on both sensor characteristics and the mechanisms through which they detect fire effects. Differences in revisit frequency, for example, influence how quickly observations can be acquired after fire (Table 1). Multi-spectral sensors, including Landsat, Sentinel-2, and HLS, estimate burn severity using changes in spectral indices such as the Normalized Burn Ratio (NBR), which is sensitive to fire-induced changes in surface reflectance (Key and Benson 2006; Miller and Thode 2007). However, these observations are strongly affected by clouds, aerosols (i.e., atmospheric dust and smoke), snow, and ice, all of which can limit the detection of vegetation and soil conditions (Melchiorre et al. 2020; Holsinger et al. 2022; Tang et al. 2023). In contrast, SAR systems such as Sentinel-1 actively transmit and receive microwave energy that is sensitive to vegetation structure and moisture (Kasischke et al. 1997). Because microwave signals can penetrate most atmospheric conditions, SAR observations may be collected regardless of cloud cover or aerosol loading (Carreiras et al. 2020; Reiche et al. 2021; Silva et al. 2022). Although snow, ice, and surface moisture can affect some applications, SAR data can still be useful for characterizing vegetation under limited snow cover (e.g., Bucha et al. 2021). SAR-based burn-severity mapping typically relies on fire-induced changes in backscatter intensity, with co- and cross-polarized channels emphasizing different surface properties (Tarazona et al. 2025). While multispectral data may correspond more closely with field-based measures of burn severity (Tanase et al. 2015; Zhang et al. 2024) SAR observations may substantially reduce delays in cloudy regions. Together, these differences suggest important trade-offs among data sources used for post-fire assessment.

Given their lower revisit times and (for Sentinel-1) differing sensitivities to atmospheric and surface conditions, HLS, Sentinel-1, and Sentinel-2 may reduce observational latency relative to Landsat; however, their ability to accurately characterize burn severity shortly after fire remains poorly understood. Previous studies have largely focused on ‘extended assessments’ from imagery collected several months to a year after fire (Miller and Thode 2007; Parks et al. 2014; Howe et al. 2022). Although these extended assessments support long-term planning, they do not address the needs of immediate post-fire management. Moreover, land-surface conditions change rapidly during the first year after fire (Miller and Quayle 2015). Delayed tree mortality can alter vegetation structure in forested ecosystems (Hood et al. 2018; Busby et al. 2024; Saberi et al. 2026), while rapid understory regeneration may obscure overstory damage (Holsinger et al. 2022; Reilly et al. 2023). Post-fire management actions, such as salvage logging, can dramatically alter

surface characteristics (Zuspan et al. 2024). Burn-severity indices are also affected by ash and charcoal, which can change substantially within the few months after burning (Lewis et al. 2021). Prior evaluations of early post-fire assessments have generally been limited to comparisons between Landsat and Sentinel-2 (Dannehyrolles et al. 2024) or among processing approaches within a single sensor (Vanderhoof et al. 2025). To our knowledge, no study has comprehensively evaluated these commonly used data sources—HLS, Landsat, Sentinel-1, and Sentinel-2—for early burn-severity assessment. Together, these uncertainties highlight a clear research need: to better understand how acquisition timing and remote sensing data selection affect the reliability of early burn-severity assessments.

In this study, we evaluate the availability and performance of freely available, moderate-resolution satellite datasets for early post-fire assessments in Alaska, the contiguous US, and Canada. We first quantify the extent to which aerosols, clouds, and snow/ice obstructed Earth observations following 9,129 large fires (i.e., ≥ 404 ha) that occurred from 2016 to 2024 (*Objective 1*). Second, we compare observational latency, data coverage, and post-fire acquisition characteristics among HLS, Landsat, Sentinel-1, and Sentinel-2 (*Objective 2*). Finally, using a subset of 77 fires with field-derived burn-severity measurements collected 0 to 3 years after fire, we evaluate how well burn-severity maps produced from each dataset correspond to on-the-ground conditions and how accuracy varies with acquisition timing (*Objective 3*). Together, these analyses provide a broad-scale evaluation of satellite data availability and performance, informing data selection and workflow development for operational post-fire assessment.

Methods

Study area

Our ~19.1 M km² study area spans Alaska, the contiguous US, and Canada, encompassing all major Köppen-Geiger climate zones (Beck et al. 2018). Disturbance regimes vary substantially throughout this region (Acil et al. 2024; Qiu et al. 2025). Fire size, frequency, and severity are shaped by interacting top-down (e.g., climate) and bottom-up drivers (e.g., vegetation, land use, topography) that differ across Canada and the US (Krawchuk et al. 2009; Balch et al. 2017; Mahood et al. 2025). Fire seasonality across the study area is bimodal, with distinct peaks in the spring and summer months (Fig. S1). To evaluate data obstruction (*Objective 1*) and observational latency and coverage (*Objective 2*), we analyzed large fire events across multiple ecosystem types (e.g., forests, shrublands, grasslands). For analyses comparing remotely sensed datasets with field-based burn severity

indicators (*Objective 3*), we focused on forested ecosystems (~34% of the land area of the US; ~42% of Canada; Ritchie 2021), where field data were most consistently available. Forests of western Canada and the western US are dominated by obligate-seeding evergreen conifers (Agee 1996; Baker 2009), though broadleaf species such as quaking aspen (*Populus tremuloides*) are also important in some areas. Broadleaf deciduous and mixed broadleaf-conifer forests are common throughout eastern Canada and the eastern US (Homer et al. 2020; Hermosilla et al. 2022).

Data, objectives 1–3 – fire perimeters

We compiled fire data from Canada's National Burned Area Composite (NBAC) and the US Monitoring Trends in Burn Severity (MTBS) programs, both of which integrate satellite observations and manual fire delineation to describe the location and timing of individual fire events (Eidenshink et al. 2007; Canadian Forest Service 2024). To standardize datasets, we retained only fires ≥ 404 ha, the minimum fire size mapped throughout the western US (smaller fires are included in eastern US and Canadian datasets). We also removed cross-border duplicates and restricted analyses to fires that ignited between January 2016 and December 2024 – the period with consistent Landsat, Sentinel-1, and Sentinel-2 coverage. Although several satellite launches and sensor failures occurred during this period, coverage remained sufficiently consistent for comparison. Our final sample included 9,129 large fires. Because MTBS had not yet mapped some 2023–2024 US fires at the time of data acquisition (i.e., October 12th, 2025), a small number of fires were likely omitted. For analyses of field-based burn severity assessments (*Objective 3*), we incorporated an additional 15 fires not included in MTBS using the Treatment and Wildfire Interagency Geodatabase (Call et al. 2025). We assigned end dates to each fire using thermal anomalies (i.e., 'hotspots') derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) and the Visible Infrared Imaging Radiometer Suite (VIIRS) (Giglio et al. 2016; National Aeronautics and Space Administration 2025), identifying the final hotspot date within 1 km of the fire perimeter (Parks 2014) and within 6 months of ignition. Although they may differ from agency-defined containment, these dates better represent when fire activity subsides and post-fire planning can begin. For the 6.4% of fires without detected hotspots, we assigned a 30-day duration, the mean across all events. We note that "overwintering" fires sometimes occur in boreal systems (Baltzer et al. 2025), which may have led to underestimates of the duration of some high-latitude fires. We processed fire data using the *terra* (v. 1.8–29; Hijmans

2025) and *sf* (v. 1.0–14; Pebesma 2025) packages in R (v. 4.3.1; R Core Team 2023).

Data, objectives 1–3 – remotely sensed datasets and processing

We acquired and processed all data for burn-severity mapping within Google Earth Engine (GEE) (Gorelick et al. 2017). First, we obtained surface reflectance imagery from version 2.0 of the Harmonized Landsat Sentinel-2 (HLS) dataset (Claverie et al. 2018; Ju et al. 2025), which provides Landsat OLI and Sentinel-2 data co-registered to a common 30-m grid with consistent geometric, radiometric, and atmospheric correction. This harmonization ensures that differences in latency, coverage, or accuracy reflect true sensor characteristics rather than differences in processing workflows. From this dataset, we compared Landsat-only (HLSL30), Sentinel-2-only (HLS20), and merged HLS products (including imagery from both platforms) in our analyses. For each Landsat or Sentinel-2 scene, we excluded areas with clouds, adjacent clouds, cloud shadows, or snow/ice using the Fmask-derived quality assurance (QA) band (Qiu et al. 2019), and masked pixels that were outside of the reflectance range (0,1) (i.e., under or oversaturated). We did not use the QA flag to mask water because we found that this often removed high-severity areas in post-fire imagery; instead, we used a global land cover mask to exclude perennial surface water in all scenes (Hansen et al. 2013). We also acquired Sentinel-1 ground-range-detected scenes collected at 10-m-pixel spacing in interferometric wide swath mode and dual polarization. We generated analysis-ready Sentinel-1 data products by applying border noise correction, speckle filtering, and terrain normalization using established GEE workflows (Mullissa et al. 2021). We then aggregated Sentinel-1 data to the 30-m HLS grid using the mean value of all 10-m pixels within each 30-m footprint. Accordingly, all datasets (HLS, Landsat, Sentinel-1, and Sentinel-2) were analyzed at 30-m resolution. For each Landsat, Sentinel-1, and Sentinel-2 scene, we obtained the timestamp of when the observation was acquired by the sensor for subsequent analyses. Therefore, observational latency reflects data acquisition only and excludes processing and delivery time, which may change due to shifts in technology and processing methods.

Using Landsat, Sentinel-2, and HLS data, we calculated the relativized burn ratio (RBR) as an index of burn severity (Parks et al. 2014). RBR is derived from the differenced normalized burn ratio (dNBR), which contrasts information in near-infrared and shortwave-infrared wavelengths to detect fire-related change (Key and Benson 2006). As a relativized index, RBR corrects dNBR for differences in pre-fire vegetation cover and phenology,

enabling more consistent comparisons across vegetation types and years (Miller and Thode 2007). Based on correlations with in-situ field data, RBR has comparable accuracy to other commonly used multispectral burn severity indices (Howe et al. 2022). We calculated RBR as follows:

$$NBR = (NIR - SWIR2)/(NIR + SWIR2) \quad (1)$$

$$dNBR = (NBR_{prefire} - NBR_{postfire}) \times 1000 \quad (2)$$

$$RBR = dNBR/(NBR_{prefire}) + 1.001 \quad (3)$$

Where, in Eq. 1, NIR represents B5 (Landsat) or B8A (Sentinel-2), and SWIR2 represents B7 (Landsat) or B12 (Sentinel-2).

Using Sentinel-1, we calculated the normalized radar burn ratio (nRBR) (Tarazona et al. 2025). The nRBR integrates an expected post-fire increase in co-polarized (VV) backscatter and decrease in cross-polarized (VH) backscatter into a single index (Zhou et al. 2019; De Luca et al. 2022). We computed nRBR as follows:

$$BR = \gamma^0_{postfire}/\gamma^0_{prefire} \quad (4)$$

$$nRBR = (BR_{VH} - BR_{VV})/(BR_{VH} + BR_{VV}) \times 1000 \quad (5)$$

Where γ^0 in Eq. 4 represents the normalized backscatter coefficient in VV or VH polarization on a linear-scale (Fernandez-Carrillo et al. 2019).

Data, objective 3 – field-derived indicators of burn severity

We compiled field-derived burn-severity assessments across the United States and Canada (Table 2; Fig. S2), focusing on measurements of the Composite Burn Index (CBI), basal area (BA) loss, and tree mortality to represent a range of fire effects in forested systems. CBI characterizes changes in soil, vegetation, and fuel characteristics into a single value ranging from 0 to 3 (Key and Benson 2006). BA loss and tree mortality quantify the proportion of live trees killed by fire, with BA loss emphasizing larger trees and mortality emphasizing numerically abundant smaller trees. We compiled field datasets from previously published and ongoing studies designed independently by individual investigators (Table 2). While plot designs and site selection varied among studies, they were all established to monitor fire-caused changes in forest systems. Plot sizes ranged from <0.01 to 0.1 ha, but most (82.2%) corresponded to a 30-m satellite pixel footprint (0.04–0.09 ha). We extracted remotely sensed metrics using the area-weighted mean of all 30-m pixels within each plot footprint. We accounted for minor differences in plot layout and study design in subsequent analyses (*Analysis – Objective 3*). Field plots were representative

of broader patterns of burn severity across the study area but were disproportionately sampled from coniferous forests in the western US (Fig. S1).

For our analyses, we retained plots that: (1) were within fires that burned between 2016 and 2024, the common period of availability across remotely sensed datasets; (2) contained live trees in the year before a fire, based on forest inventory records, CBI strata data (when available), or annual land cover maps (Allred et al. 2021; Hermonilla et al. 2022); (3) had valid coordinates within the fire perimeter (or within 1 km for a small number of unburned control plots); and (4) were ≥ 30 m from the nearest plot. In cases where multiple plots were within 30 m of one another, we randomly selected one to be retained. The final dataset included 1,675 plots with CBI (45 fires), 1,442 plots with BA loss (55 fires), and 1,377 plots with tree mortality (55 fires). In total, this dataset includes 2,714 unique field plots from 77 fire events. The timing of plot collection varied: 9.0% immediately post-fire, 63.7% one-year post-fire, 20.9% two years post-fire, and 6.4% three years post-fire. Because differences in the timing of plot collection may influence correspondence with post-fire imagery, we accounted for these differences in subsequent analyses (*Analysis – Objective 3*).

Analysis, objective 1 – obstructions due to aerosols, clouds, and snow

We quantified broad-scale obstruction by aerosols, clouds, and snow/ice throughout our study area using daily MODIS and VIIRS products, which provide a reasonable proxy for other, higher-spatial-resolution data sources (Melchiorre et al. 2020). To quantify aerosol loading, we used the 1-km MODIS MCD19A2 (V6.1) aerosol optical depth (AOD) dataset (Lyapustin and Wang 2022) in blue wavelengths (0.47 μ m), which are sensitive to smoke and other small particulates. We classified pixels as aerosol-obstructed if $AOD \geq 0.5$, a common threshold for heavy aerosol loading (e.g., Kahn and Gaitley 2015). We identified cloud obstruction using quality flags in the 1-km VIIRS VNP09GA surface reflectance product (Vermote et al. 2023), designating pixels as obstructed when flagged as probably cloudy, confidently cloudy, or containing cloud shadows. For snow and ice, we used the 500-m MODIS MOD10A1 (V6.1) snow-cover product (Hall and Riggs 2021), aggregated using the mean value in a 1-km pixel, and marked pixels as obstructed when snow cover was >0 . For each dataset, we calculated the percentage of obstructed days for each region from October 1st, 2015, through September 30th, 2025 (water years 2016–2025). We also assessed post-fire obstruction during the first month after large fires (2016–2024) by computing the mean proportion of obstructed days across all pixels within each fire perimeter.

Table 2 A summary of field data used for validation of remotely sensed burn-severity assessments throughout Canada and the US. The locations of field sites are shown in Fig. S2

Contributor(s)	CBI – fires (plots)	BA loss – fires (plots)	Mortality – fires (plots)	Years since fire	Study area	Reference(s)
Barrett, K. J., Stevens-Rumann, C. S., and Weimer, K	0 (0)	4 (134)	4 (134)	1–3	Western US	(Weimer et al. In Review)
Beinhart, C. A., Dappen, P. R., Mahan, K. R., Tomczyk, N. and Yannayon, E	0 (0)	2 (277)	2 (277)	0–2	Western US	(Beinhart and Mahan 2026)
Boucher, Y., and Danneyrolles, D	4 (193)	0 (0)	0 (0)	0–1	Eastern Canada	(Danneyrolles et al. 2024)
Clark-Wolf, K. D	0 (0)	2 (49)	2 (49)	3	Western US	(Clark-Wolf et al. 2022 ; Davis et al. 2023)
Crouse, J. E., Roccaforte, J. P., and Stoddard, M. T	0 (0)	3 (80)	3 (80)	1–3	Western US	(Stoddard et al. 2024 , 2021 , Unpublished Data)
Corbett, M. J. and James, P. M. A	3 (300)	0 (0)	0 (0)	1	Eastern Canada	Unpublished Data
Davis, K. T	0 (0)	2 (25)	2 (27)	2–3	Western US	(Davis et al. 2023)
Harvey, B. J., and Saberi, S. J	14 (297)	14 (298)	14 (298)	1	Western US	(Saberi et al. 2022 ; Howe et al. 2022)
Harvey, B. J., Laughlin, M.M., and Morris, J. E	3 (56)	3 (56)	3 (45)	2	Western US	(Morris et al. 2025 , Unpublished Data)
Hood, S. M., and Carrasco, J. T	0 (0)	8 (80)	8 (80)	0–2	Western US	(Cansler et al. 2020 , Unpublished Data)
Howe, A. A	10 (316)	0 (0)	0 (0)	1	Western Canada, Eastern US, Western US	(Howe et al. 2022 ; Parks et al. 2019 ; Picotte, et al., 2019 , Unpublished Data)
Iniguez, J. M., and Rodman, K. C	2 (41)	0 (0)	0 (0)	1	Western US	Unpublished Data
Kreider, M. R. and Yocom, L. L	0 (0)	16 (394)	16 (338)	2–3	Western US	(Kreider and Yocom 2021a, b ; Davis et al. 2023)
Parisien, M.-A., and Whitman, E	5 (187)	0 (0)	0 (0)	1	Western Canada	(Dawe et al. 2025 ; Parks et al. 2019 ; Whitman et al. 2020 , Unpublished Data)
Rodman, K.C	0 (0)	3 (19)	3 (19)	1–2	Western US	(Rodman et al. 2025b)
Stevens-Rumann, C. S., and Woollet, J. N	0 (0)	1 (12)	1 (12)	1	Western US	(Woollet et al. 2023)
Stevens-Rumann, C. S., and Wooten, J. T	0 (0)	1 (18)	1 (18)	2	Western US	(Wooten et al. 2022 ; Davis et al. 2023)
Twaddell, E. M. N., and Vorster, A. G	2 (154)	0 (0)	0 (0)	1	Western US	(Twaddell 2023 ; Vorster et al. 2024)
Woodward, B. D	1 (23)	0 (0)	0 (0)	1	Western US	(Woodward et al. 2020)
Yocom, L. L	2 (108)	0 (0)	0 (0)	1	Western US	Unpublished Data
Total	45 (1675)	55 (1442)	55 (1377)	0–3		

To illustrate how these obstructions can affect post-fire burn-severity assessment in individual events, we characterized temporal patterns of fire spread and data obstruction during the 2020 Bush Fire (Arizona, USA) and 2020 Cameron Peak Fire (Colorado, USA). These two events span the typical range of conditions under which large fires might burn in our study area, ranging from relatively simple (Bush Fire) to very challenging (Cameron Peak) to map. For example, the Bush Fire ignited on June 13th and was contained by July 6th, and relatively rapid

acquisition of post-fire satellite imagery. In contrast, the Cameron Peak fire ignited on August 13th, wasn't fully contained until December 2nd, and had no clear post-fire scenes from Landsat or Sentinel-2 for more than six months after the fire. Using the 30-m, daily fire-spread reconstruction approach of Parks ([2014](#)), we mapped the progression of each fire and paired this information with daily aerosol, cloud, and snow obstruction estimates derived from the datasets above, for a six-month period beginning in the month of each fire ignition.

Analysis, objective 2 – observational latency and data coverage following large fires

Using each fire's end date, we calculated the number of days required to obtain the first usable scene from HLS, Landsat, Sentinel-1, and Sentinel-2. For HLS, Landsat, and Sentinel-2, we defined a usable scene as one with at least 80% of the fire perimeter covered by unobstructed pixels – free from clouds, cloud shadows, snow/ice, or high aerosol loading in the QA band (Qiu et al. 2019). For Sentinel-1, which is less affected by these obstructions, we defined a usable scene as one with at least 80% overlap between the fire perimeter and footprint of valid data. Pixels with perennial surface water (Hansen et al. 2013) were excluded from this calculation for all datasets. This threshold quantifies the ability of an analyst to use a 'paired-scene' approach with a relatively clear image. Higher thresholds for the unobstructed proportion of a fire perimeter, while qualitatively similar, led to many fires and several regions without any usable scenes. We also summarized early data coverage by calculating a composite image using the pixel-wise mean of all unobstructed pixels from all scenes acquired in the first post-fire month and determining the proportion of each fire covered by the composite. We compared sensors by examining temporal (median days to a usable scene by fire year, median composite coverage) and spatial patterns (median values by Level III ecoregion).

To quantify differences in observational latency and data coverage among sensors, we fit generalized linear mixed models (GLMMs) of the form $response \sim dataset + (1|Ecoregion/Fire\ ID) + (1|Fire\ Year)$ using the *glmmTMB* package (v. 1.1.8; Brooks et al. 2017) in R. At the fire-scale, we modeled days to the first usable scene (response) with dataset (HLS, Landsat, Sentinel-1, Sentinel-2) as a fixed-effect term (predictor), using a generalized Poisson error structure. We modeled the proportion of each fire covered by the one-month mean composite using ordered beta regression, appropriate for bounded continuous data (Kubinec 2023). Both models included fire year as a random intercept term to account for variation in data availability over time, and fire ID nested within ecoregion to account for spatial dependence and regional differences in weather patterns and atmospheric conditions. We conducted post-hoc, pairwise comparisons among sensors using the *emmeans* package (v. 1.8.9; Lenth 2023), with multivariate-t adjustments for multiple comparisons. Although simpler descriptive statistics might be used to summarize these broad patterns, our objective was to formally compare datasets while accounting for strong spatial and temporal dependence among observations and other eccentricities in the data. GLMMs allowed us to estimate average differences

among sensors while accounting for patterns of dependence and a complex data structure.

Analysis, objective 3 – accuracy relative to field measures

Next, we evaluated correspondence between each remotely sensed dataset and field-derived indicator of burn severity (Table 2). To do so, we created mean composite images from HLS, Landsat, Sentinel-1, and Sentinel-2 at monthly intervals between 1–9 months after fire end dates. We calculated RBR (HLS, Landsat, and Sentinel-2) or nRBR (Sentinel-1) by comparing each post-fire composite to imagery from a similar phenological window (i.e., composite period $\pm$ two months) in the previous year. We then extracted all values (36 combinations: 9 time steps $\times$ 4 datasets) at field-plot locations. We imputed missing values in remotely sensed data due to obstructions using 100 iterations of predictive mean matching with the *mice* package (v. 3.17.0; Van Buuren and Groothuis-Oudshoorn 2011) in R, which preserves data distributions and helps to propagate imputation uncertainty into subsequent accuracy estimates.

For CBI, BA loss, and tree mortality, we fit GLMMs of the form $response \sim index \times field\ collection\ timing + (1|FireID) + (1|Contributor)$ for each response variable, dataset (HLS, Landsat, Sentinel-1, and Sentinel-2), and timestep (months after fire in which remotely sensed data were acquired), yielding 108 individual models (three responses, four datasets, and nine time steps). We modeled all responses using ordered beta regression, linearly scaling CBI from [0,3] to [0,1] prior to model fitting. Ecoregion-level and fire-year-level random effects were excluded from these models due to near-zero variance and convergence failure. However, we included random intercept terms of fire ID and contributor to account for spatial dependence and slight variations in collection protocols among field datasets, respectively. To propagate imputation uncertainty, we fit the models above, predicted plot-level responses using fixed effects (i.e., assuming mean values of random effects), and summarized model accuracy using the squared Pearson correlation (R^2) between observed and predicted values for each of the 100 imputation iterations. We also calculated balanced accuracy by classifying predictions into high-severity (near-total canopy loss; $CBI \geq 2.25$ or BA loss/mortality $\geq 95\%$) and unburned-to-moderate severity ($CBI < 2.25$ or BA loss/mortality $< 95\%$) using the index threshold that maximized the true skill statistic in each model (Allouche et al. 2006). To compare post-fire assessments with extended assessments utilizing a full growing season of imagery after the fire, we repeated these analyses using the mean-composite RBR method (Parks et al. 2018).

Results

Objective 1 – common sources of data obstruction

Data obstruction varied substantially across Alaska, the

contiguous US, and Canada, with clouds representing the most consistent limitation for multispectral imagery (Fig. 1). Aerosol obstruction was generally low (Fig. 1a)

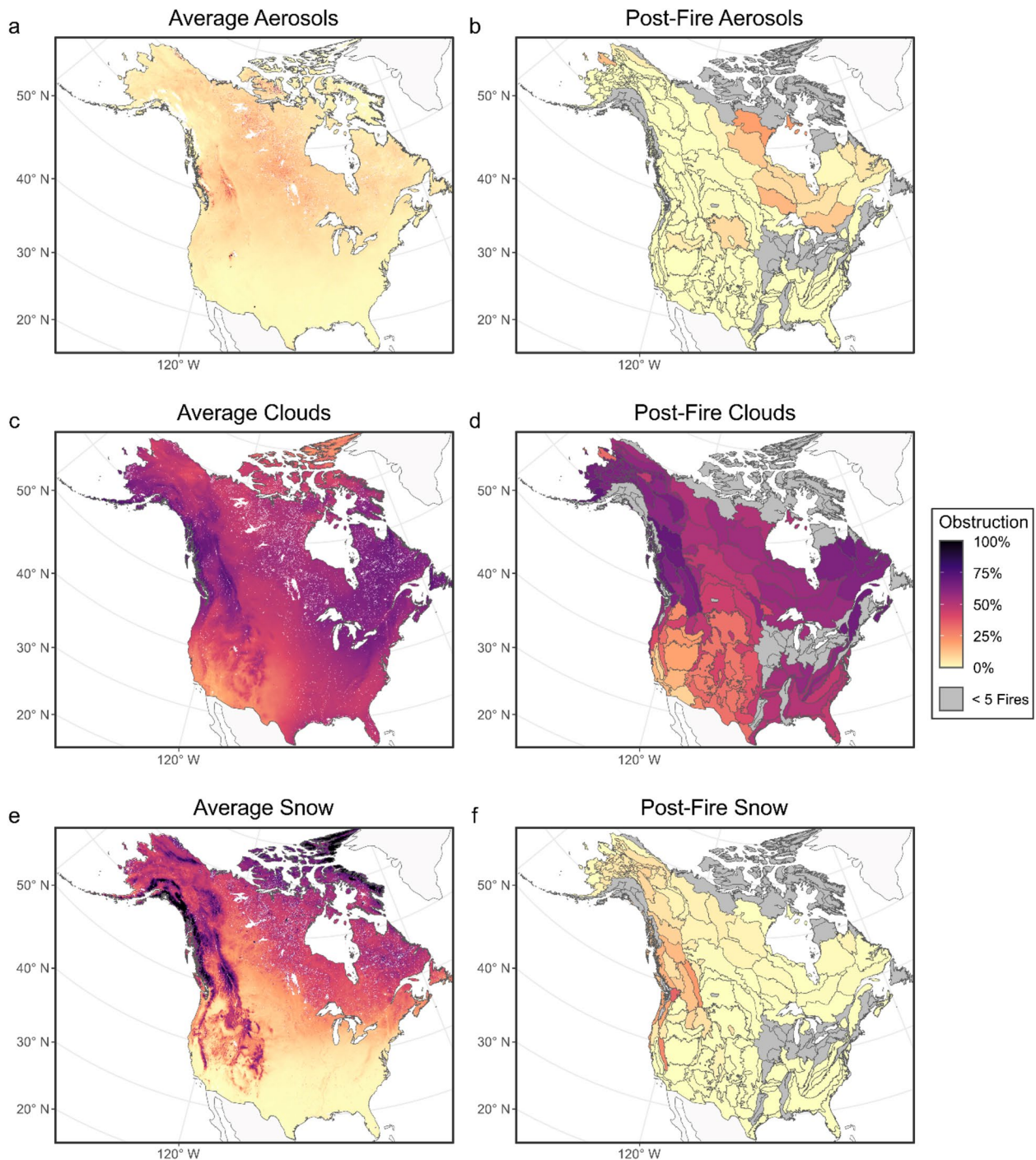


Fig. 1 Sources of remotely sensed data obstruction in Alaska, the contiguous United States, and Canada. Panels show the percentage of days with (a, b) high aerosol loading, (c, d) cloud or cloud-shadow obstruction, and (e, f) snow or ice cover. a, c, e summarize average per-pixel (1 km) obstruction during water years 2016–2025. b, d, f show obstruction in the first month after large fires (≥ 404 ha) from 2016–2024, where colors indicate median fire-level obstruction within each Level III ecoregion (CEC, 2025)

and only affected a small subset of events – 2.7% of large fires experienced heavy (>50% of days) post-fire aerosol cover, mainly in the Boreal Shield and Hudson Bay regions of eastern Canada (Fig. 1b). In contrast, cloud cover was widespread (Fig. 1c): 47.3% of fires had heavy cloud obstruction during the first post-fire month, particularly in Alaska, coastal Canada, and the southeastern US, while the western US remained relatively clear (Fig. 1d). Snow cover showed strong geographic gradients, increasing with elevation and latitude (Fig. 1e), but heavily obstructed only 3.0% of large fires. Post-fire snow

was most problematic in the Sierra Nevada, North Cascades, and Canadian Rocky Mountains (Fig. 1f).

The 2020 Bush Fire (Arizona, USA) and 2020 Cameron Peak Fire (Colorado, USA) illustrate the broad range of conditions encountered during operational fire mapping (Fig. 2). The Bush Fire was active for approximately one week in June (Fig. 2a, c), whereas the Cameron Peak Fire experienced multiple periods of spread between August and October (Fig. 2b, d). Following the Bush Fire, post-fire imagery was minimally affected by aerosols, clouds, or snow/ice. During the first month after the fire ended

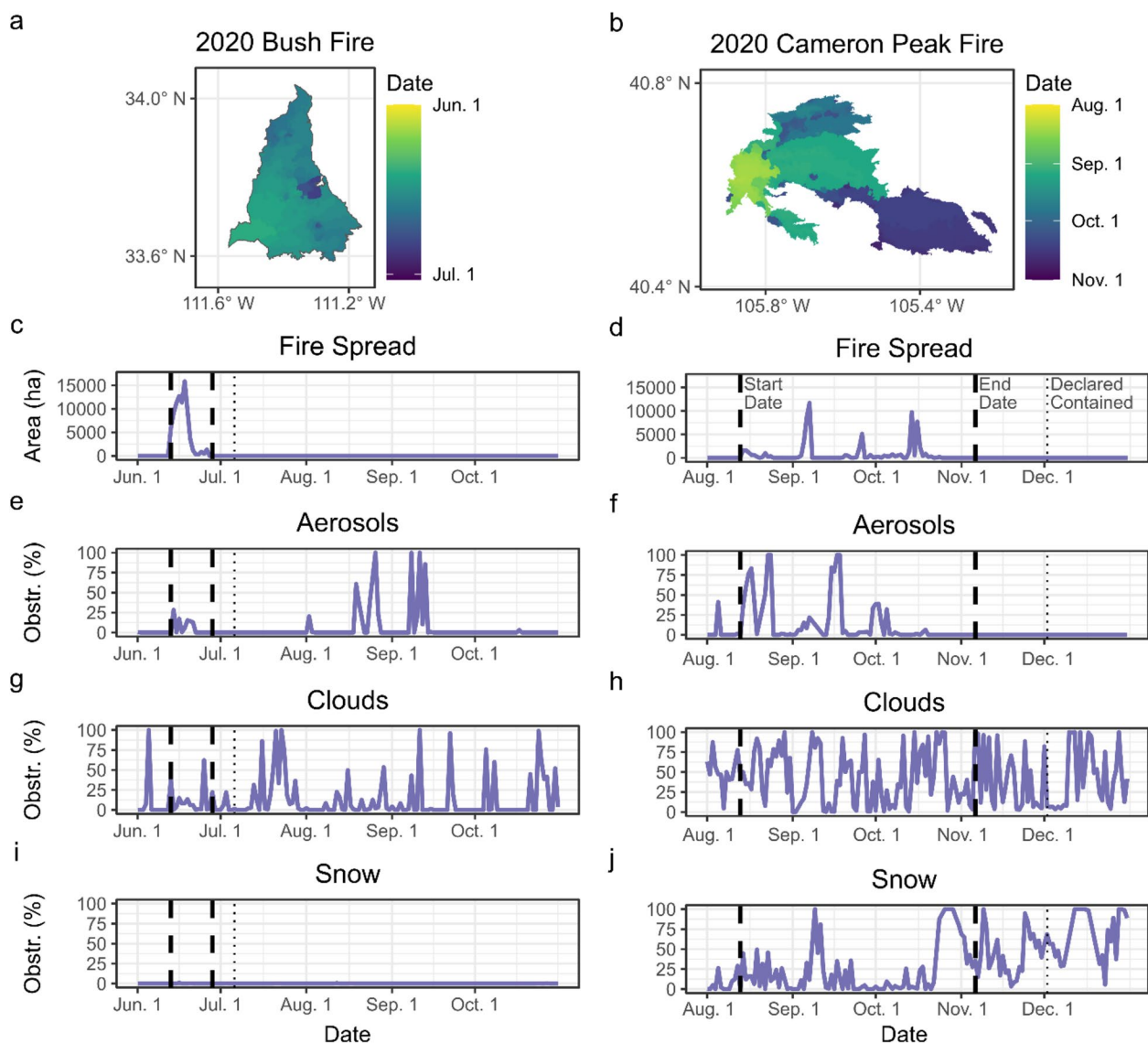


Fig. 2 Fire progression and sources of image obstruction during and after the 2020 Bush Fire (left) and 2020 Cameron Peak fire (right) in central Arizona, USA and northern Colorado, USA, respectively. **a–d** show daily fire spread, reconstructed using the methods of Parks (2014). **e–j** show the percentage of the final fire perimeter obstructed by aerosols, clouds, and snow. Nearby large fires likely contributed to elevated aerosol loading; the Superstition Fire occurred in August to the south of the Bush Fire, and several fires occurred near Cameron Peak between August and November

(June 28th), no days were heavily obstructed ($\geq 50\%$) by aerosols or snow, and clouds obstructed only 17.2% of days (Fig. 2e, g, i), suggesting that usable imagery from most sensors was available shortly after burning ceased. In contrast, post-fire assessment of the Cameron Peak Fire was substantially hindered by snow. Because the fire burned into winter and at higher elevations, snow accumulated soon after containment and heavily obstructed 55.2% of days during the first post-fire month (Fig. 2j). Clouds heavily obstructed an additional 34.5% of days, whereas aerosols caused no heavy obstruction. Though Sentinel-1 provided a usable image immediately after fire activity in Cameron Peak subsided, Landsat, Sentinel-2, and HLS did not provide an unobstructed scene for more than six months.

Objective 2 – observational latency and data coverage following large fires

Observational latency and post-fire data coverage differed among HLS, Landsat, Sentinel-1, and Sentinel-2

across the study area (Fig. 3, Table 3). Most fires had at least one usable scene ($>80\%$ unobstructed coverage) within the first post-fire month, ranging from 68.6% for Landsat to 89.2% for Sentinel-1 (Table 3). Median latency was 17 days for Landsat, 10 days for Sentinel-2, 7 days for HLS, and 4 days for Sentinel-1 (Fig. 3a; Tables S1–S3). On average, one-month composite imagery provided high post-fire coverage across datasets (Figs. S3, S4), with median fire-level coverage of 99.8% for Landsat and 100% for HLS, Sentinel-1, and Sentinel-2 (Fig. S3a; Table 3). However, the proportion of fires with near-complete ($>99\%$) coverage varied substantially, ranging from 56.1% for Landsat to 89.6% for Sentinel-1, with intermediate values for Sentinel-2 (70.5%) and HLS (82.8%) (Table 3). Similarly, the proportion of total burned area (across all fires) covered by one-month composites was 68.0% for Landsat, 81.4% for Sentinel-2, 88.0% for HLS, and 89.5% for Sentinel-1 (Table 3).

Latency and coverage also varied over time and among ecoregions due to changes in satellite missions, fire

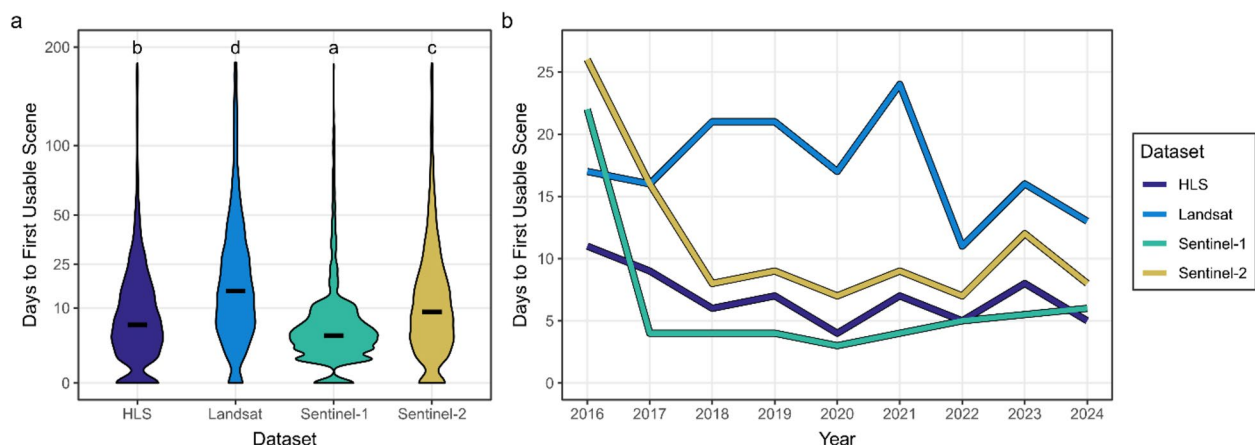


Fig. 3 Observational latency for four remotely sensed datasets collected after large fires (≥ 404 ha) across Alaska, the contiguous US, and Canada from 2016–2024. Latency represents the number of days between the fire end date and the first scene with $\geq 80\%$ unobstructed coverage of the fire perimeter (i.e., “first usable scene”). In panel (a), letters denote significance groups from a generalized linear mixed model with post-hoc pairwise comparisons ($\alpha=0.05$). b shows median latencies by fire year. The y-axis in (a) is square-root transformed to emphasize lower, data-dense values. In (b), steep declines in latency of Sentinel-1 (2017) and Sentinel-2 (2018) relate to the launch of Sentinel-1B and Sentinel-2B, respectively. Excluding fires before 2018 had little impact on results in (a). HLS: Harmonized Landsat Sentinel-2

Table 3 Observational latency and data coverage for four datasets collected after large fires (≥ 404 ha) across Alaska, the contiguous US, and Canada from 2016–2024. Usable scenes are defined as those with at least 80% clear coverage within the fire perimeter, and one-month composites combine all scenes acquired within one month of a fire end date. HLS: Harmonized Landsat Sentinel-2

Dataset	First usable scene (median)	No. fires with usable scene within one month	One-month composite coverage (median)	No. fires with full one-month coverage	Total fire area with one-month coverage
HLS	7 days	88.6%	100%	82.8%	88.0%
Landsat OLI	17 days	68.6%	99.8%	56.1%	68.0%
Sentinel-1	4 days	89.2%	100%	89.6%	89.5%
Sentinel-2	10 days	80.0%	100%	70.5%	81.4%

seasonality, and atmospheric conditions (Fig. 3b; Fig. 4; Fig. S3b). Sentinel-1 and Sentinel-2 latencies declined substantially in 2017–2018 following the launches of Sentinel-1B and Sentinel-2B, respectively. Landsat latency decreased after the launch of Landsat 9 in 2021. Conversely, Sentinel-1 latency increased modestly following the failure of Sentinel-1B in December 2021. Acquiring clear post-fire imagery was generally easiest in the southwestern US (Fig. 4), where cloud cover is low throughout much of the year (Fig. 1b). Across most ecoregions, HLS and Sentinel-1 consistently reduced time to the first usable scene relative to Landsat and Sentinel-2, with similar improvements in one-month composite coverage (Fig. 4; Fig. S4). These patterns did not change when early years of the Sentinel program were excluded (Fig. S7).

Objective 3 – accuracy relative to field-derived indicators of burn severity

HLS, Landsat, and Sentinel-2 showed similar correspondence with field-derived burn-severity indicators,

whereas Sentinel-1 performed less well overall (Fig. 5). Thus, despite providing the most rapid and complete post-fire coverage (Figs. 3, 4), early Sentinel-1 acquisitions were comparatively inaccurate. Although Sentinel-1 performance improved over time, it did not match the accuracy of multispectral datasets except during winter months. In contrast, Landsat, Sentinel-2, and HLS achieved high accuracy both shortly after fire (typically autumn in our sampled fires) and 8–9 months later (typically spring). Predictions of CBI were generally less accurate than those of BA loss and tree mortality, suggesting that fire effects on trees are easier to characterize remotely than those affecting soils and understory vegetation. As expected, the Parks et al. (2018) mean-composite approach applied to a full post-fire growing season of HLS imagery produced the strongest agreement with field data, although early post-fire assessments from HLS, Landsat, and Sentinel-2 were nearly as effective at identifying high-severity areas (Fig. 5d-f). Despite differences in performance, all datasets were strongly and positively

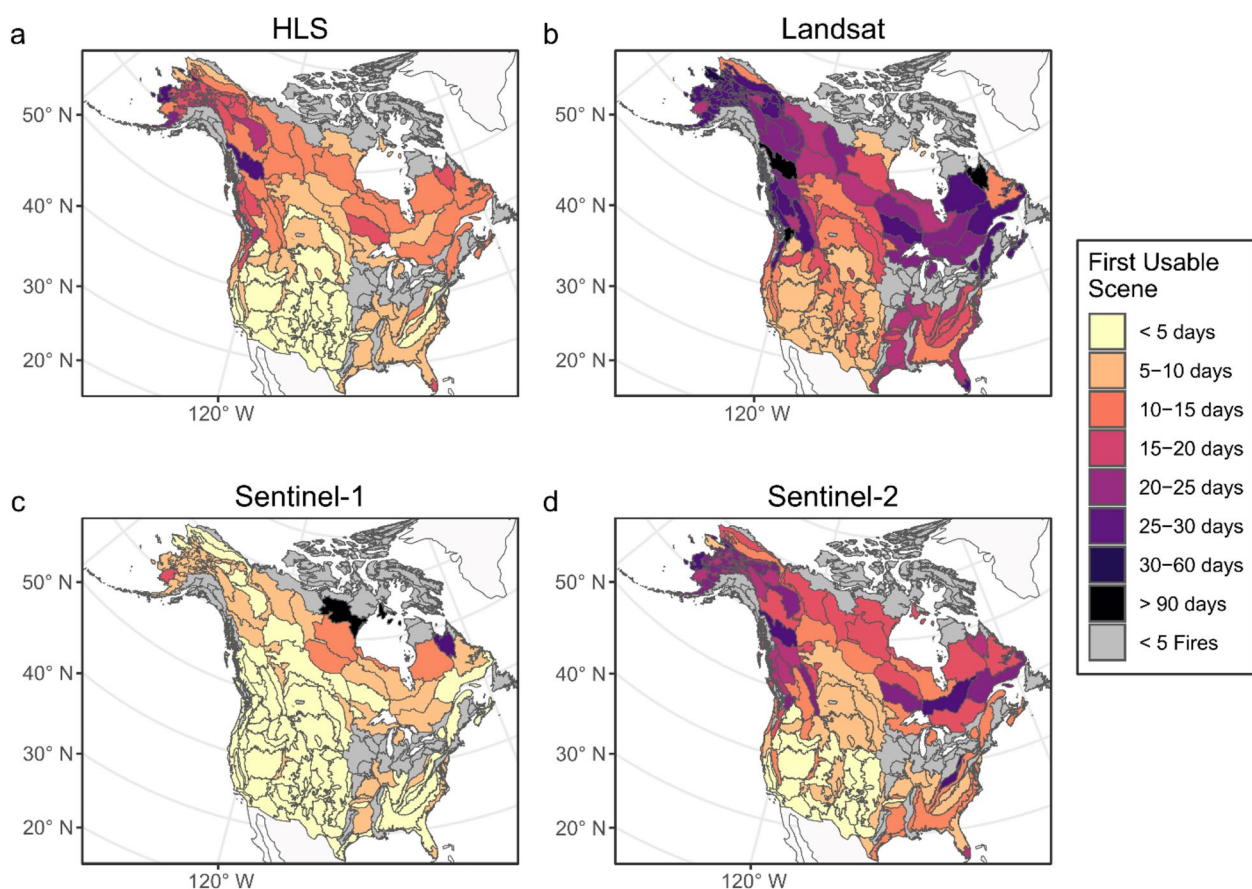


Fig. 4 Observational latency for four remotely sensed datasets (a–d) collected after large fires (≥ 404 ha) across Alaska, the contiguous US, and Canada from 2016–2024. Latency represents the number of days between the fire end date and the first scene with $\geq 80\%$ clear coverage of the fire perimeter (i.e., “first usable scene”). Map colors show median latency within each Level III ecoregion (CEC 2025). HLS: Harmonized Landsat Sentinel-2

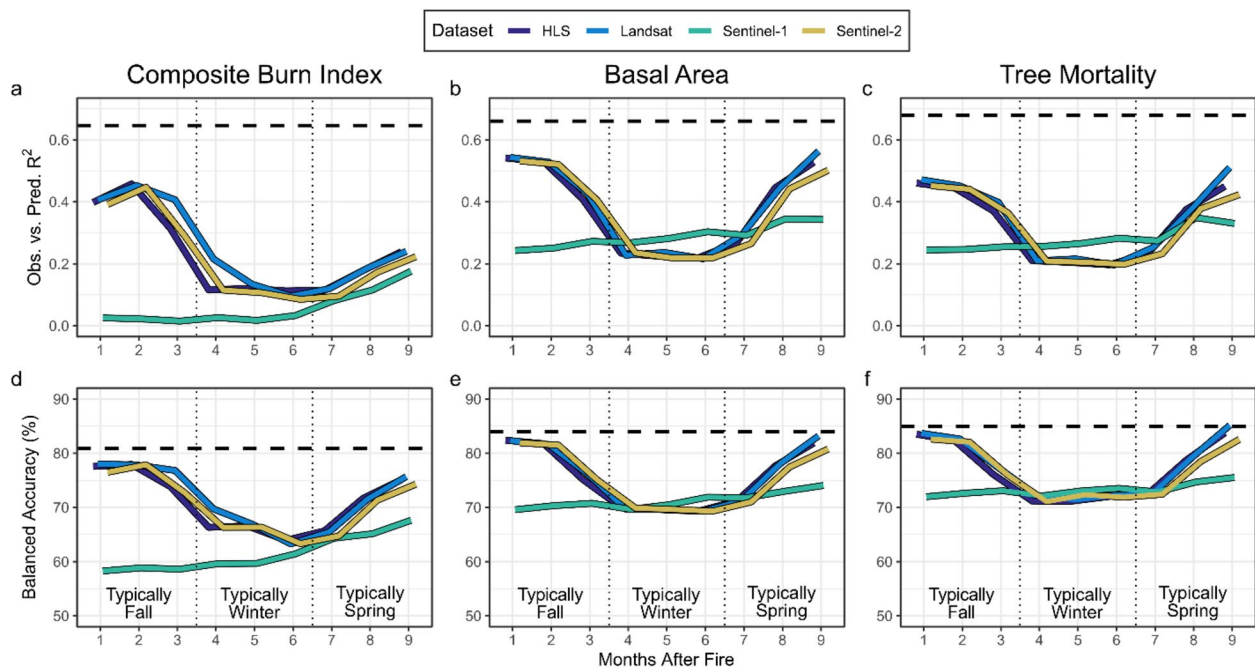


Fig. 5 Accuracy of models predicting composite burn index (CBI; **a, d**), basal area loss (**b, e**), and tree mortality (**c, f**) using remotely sensed datasets acquired at different times after fire. **d–f** show balanced accuracy for identifying plots with near-total canopy loss (CBI ≥ 2.25 or BA loss/mortality $\geq 95\%$). Accuracy values reflect fixed-effects predictions, conditional on mean values of random intercepts. Horizontal dashed lines indicate accuracy from the Parks et al. (2018) mean-composite method, which requires a full post-fire growing season of imagery. HLS: Harmonized Landsat Sentinel-2

related to burn severity, and imputation uncertainty had minimal effects on model coefficients or accuracy estimates (Tables S7–S9). Field plots measured 2–3 years after fire generally exhibited higher CBI, BA loss, and tree mortality at the same remotely sensed index value, consistent with delayed tree mortality and other longer-term fire effects not fully captured by early post-fire imagery (Fig. S5). Thresholds used to classify high-severity fire

also varied seasonally and with acquisition timing (Fig. 6), likely reflecting changes in vegetation condition as well as ash and charcoal cover.

Discussion

Timely, reliable burn-severity assessments are critical for hazard mitigation and post-fire recovery planning, yet atmospheric and surface conditions often delay or

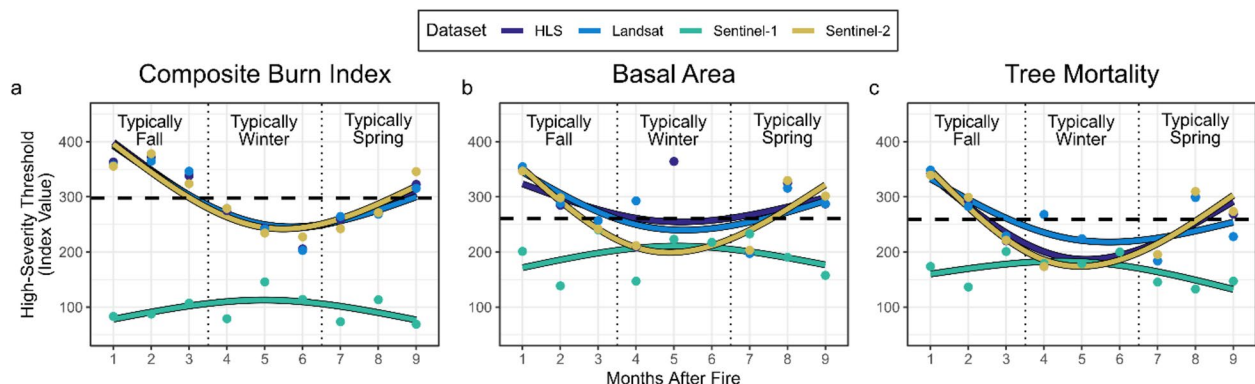


Fig. 6 Summary of thresholds used to define near-total canopy loss based on the composite burn index (**a**), basal area loss (**b**), and tree mortality (**c**) using remotely sensed datasets acquired at different times after fire. Horizontal dashed lines indicate thresholds from the Parks et al. (2018) mean-composite method, which requires a full post-fire growing season of imagery. Colored lines show the smoothed relationship between high-severity thresholds and time since fire, using a natural spline term with two degrees of freedom. HLS: Harmonized Landsat Sentinel-2

degrade remotely sensed observations. Using nearly 10,000 fires across Canada and the US, we addressed three questions relevant to operational post-fire mapping: (1) *What most commonly obstructs post-fire observations?* (2) *How quickly and completely can moderate-resolution sensors observe recently burned areas?* And (3) *How well do early remote-sensing products correspond with field-based burn-severity indicators in forests?* Cloud cover was the most common source of post-fire data obstruction faced by operational mapping programs. We also found that both SAR (Sentinel-1) and high-frequency multispectral data (HLS) substantially reduced observational latency and increased spatial coverage relative to Landsat, potentially accelerating burn-severity assessments. However, multispectral datasets produced more accurate burn-severity estimates in forested ecosystems than SAR. Together, these findings can inform operational mapping protocols and improve post-fire decision-making in North America and other regions facing increases in extreme wildfire activity.

Atmospheric conditions challenge early burn-severity assessments

We found that clouds and cloud shadows were the dominant sources of post-fire image obstruction, heavily affecting ~47% of fires in our study area (Fig. 1). As a result, they represent the primary obstacle facing operational burn-severity mapping programs that rely on multispectral imagery. This finding is consistent with previous studies documenting widespread cloud-related limitations, particularly in mesic and coastal ecoregions (Melchiorre et al. 2020; Carreiras et al. 2020; Zhang et al. 2022; Tang et al. 2023). In contrast, snow and aerosols were only important in specific locations. Post-fire snow cover was most prevalent in the Sierra Nevada, North Cascades, and Canadian Rockies ecoregions, where substantial winter and spring snowfall often follows the fire season (Eidenshink et al. 2007; Abatzoglou et al. 2018; Canadian Forest Service 2024; NASA 2025). Aerosol obstruction was concentrated around Hudson Bay and the Boreal Shield, where persistent haze is influenced by emissions from Europe and Asia (Abbatt et al. 2019). During major fire years, synchronized burning across Boreal Canada (Parisien et al. 2023) likely further increased aerosol loading in these areas through long-range smoke transport from active fires in western Canada.

Although snow and aerosols heavily obstructed a relatively small proportion of fires (~3%) compared to clouds, their importance may increase under future fire regimes. Human-caused ignitions are lengthening fire seasons into spring and autumn months (Balch et al. 2017; Parisien et al. 2023; Ellis et al. 2025), when snow is

more likely to impede post-fire assessment. In our dataset, snow occurred on just 2.9% of days following summer fires (June–August), compared with 7.8% of days following spring and autumn fires. At the same time, fire-conducive weather is becoming increasingly synchronous across North America (Abatzoglou et al. 2021), increasing the likelihood that smoke from large fires will obstruct mapping efforts far downwind, thereby hindering post-fire operations (e.g., Fig. 2e, f). Consistent with this pattern, aerosols obstructed 6.8% of days during the five largest fire years (2017, 2018, 2021, 2022, and 2023), compared with 3.2% in all other years. These findings highlight an emerging challenge that impacts post-fire assessment and other applications of remotely sensed data – as changes in climate and land use alter the processes being monitored, mapping practices may require modification and refinement to remain effective. Datasets with shorter revisit intervals, methods that are less sensitive to snow, and sensors capable of operating through atmospheric obstructions offer potential pathways to improve future post-fire assessments.

Reducing delays in operational burn-severity mapping

In our comparison of four freely available moderate-resolution datasets, we found consistent differences in observational latency and coverage among sensors, indicating that dataset selection strongly influences the timeliness of post-fire assessments (Fig. 3; Table 3). Leveraging the cloud-penetration ability of SAR as well as short revisit intervals, Sentinel-1 provided rapid post-fire coverage across Canada and the US, consistent with research on other continents (Reiche et al. 2021; Silva et al. 2022; Tang et al. 2023). HLS also exhibited low observational latency because its cross-platform harmonization increases sampling frequency and the likelihood of acquiring cloud-free imagery (Li and Roy 2017; Claverie et al. 2018). However, harmonization requires aggregating Sentinel-2 imagery to the 30-m Landsat grid, potentially reducing its utility for ecological applications that depend on fine spatial detail. For example, native 10- or 20-m Sentinel-2 imagery can better detect features such as individual trees (Wasserman et al. 2019; Howe et al. 2022), which might be missed at 30-m resolution (Chapman et al. 2020). Nevertheless, for operational applications that prioritize rapid and complete coverage of burned areas, the increased revisit frequency of HLS may outweigh this loss of spatial detail. When combined with image compositing, both Sentinel-1 and HLS achieved near-complete coverage for most fires (>80% of events) within one month of the last hotspot detection (Table 3). Annual or seasonal image composites are now commonplace in time-series analysis (Hermosilla et al. 2015; Kennedy et al. 2018; Schleeweis et al. 2020; *but see* Zhu et al. 2020),

and many extended burn-severity assessments (Whitman et al. 2020; Guindon et al. 2021). These same approaches may also improve early post-fire assessments (e.g., Holsinger et al. 2022; Vanderhoof et al. 2025). Although compositing does not significantly increase accuracy (Parks et al. 2018; Howe et al. 2022), it can increase spatial coverage by combining partially obstructed scenes also streamlining the preprocessing steps used in traditional paired-scene approaches. Taken together, our findings suggest that expanded use of SAR or high-frequency multispectral imagery (e.g., HLS), combined with compositing approaches, could substantially reduce delays in post-fire burn-severity assessments.

Multispectral imagery most strongly corresponds with field-derived burn severity

Across 77 fires spanning broad environmental gradients, multispectral imagery (HLS, Landsat, and Sentinel-2) acquired during spring and autumn months corresponded well with field-derived burn-severity indicators, outperforming winter imagery and SAR data (Fig. 5). Growing-season imagery is widely used in change-detection applications because the timing of peak photosynthetic activity accentuates the signal of vegetation change (Kennedy et al. 2018; Parks et al. 2019; Guindon et al. 2021). However, many fires require assessment outside this optimal window; in our dataset, ~23% of large fires across Alaska, Canada, and the contiguous US ended after September 1st, when snow cover and leaf-off conditions can reduce optical-sensor performance. In these cases, approaches that exploit snow cover to identify changes in evergreen vegetation may be useful (e.g., Baker et al. 2017; Rodman et al. 2021; Stueve et al. 2011; Vanderhoof et al. 2020), although they are less effective for assessing soil impacts or changes in deciduous and senescent vegetation. Alternatively, image composites spanning autumn and spring can maintain strong accuracy, improve coverage, and may be useful for operational mapping (Holsinger et al. 2022; Howe et al. 2022).

Importantly, early post-fire assessments derived from multispectral imagery identified high-severity areas nearly as well as extended assessments that use a full growing season of imagery (e.g., Parks et al. 2018; Whitman et al. 2020). For example, balanced accuracy for detecting high-severity fire ($\geq 95\%$ BA loss or tree mortality) exceeded 80% within two months of fire (Fig. 5e, f). This finding is particularly important because high-severity areas, which are largely devoid of vegetation and organic matter, are at the greatest risk of erosion, mass wasting, extreme runoff, and long-term ecosystem change. Consequently, they are often prioritized for post-fire management and rehabilitation efforts (Danneynrolles

et al. 2024; Jones et al. 2025; Robichaud et al. 2009; Rodman et al. 2025a; Stevens et al. 2021).

Beyond operational applications, our findings have important implications for ecological research. Burn-severity maps are widely used to explain patterns of post-fire regeneration, delayed tree mortality, vegetation conversion, carbon dynamics, and forest resilience at local to continental scales (Turner et al. 2019; Davis et al. 2023; Busby et al. 2024; Coop et al. 2026). However, accurately characterizing burn severity is more challenging in some regions than others. For example, fires in the southeastern US are often low-severity surface burns, and rapid vegetation regrowth can complicate remote-sensing analyses (Parks et al. 2019; Vanderhoof et al. 2025). Consistent with this pattern, model performance varied across forest types and regions in our study (Fig. S6). Thus, region-specific models that account for local vegetation characteristics, fire regimes, and seasonality will likely outperform the continental-scale models presented here.

Matching remote-sensing strategies to management objectives

Overall, patterns of observational latency, data coverage, and accuracy reveal a clear trade-off between multispectral and synthetic aperture radar (SAR) data in operational mapping efforts: SAR enables rapid post-fire assessment but with lower accuracy, whereas multispectral imagery provides higher accuracy but is often delayed by atmospheric obstruction. Combining these data sources in post-fire assessments may help leverage the strengths of each. For example, Sentinel-1 can be used to gap-fill obstructed multispectral imagery through deep-learning approaches with minimal loss of accuracy (Hu et al. 2023). Multispectral and radar data are also highly complementary and are increasingly integrated for applications such as land-cover mapping (Xu et al. 2024) suggesting that hybrid approaches could improve both the timeliness and accuracy of early post-fire assessments. However, the optimal mapping strategy ultimately depends on the intended use. For planning post-fire reforestation, which typically occurs within three years of fire (Rodman et al. 2024), delaying assessment until the following growing season may be sufficient and could improve characterization of delayed tree mortality (Busby et al. 2024) and natural tree regeneration (Holden et al. 2025). Nevertheless, some agencies conduct reforestation assessments during the first fall and winter after fire, and compositing approaches that leverage the most recent available imagery can improve map accuracy for these efforts. In contrast, applications such as landslide forecasting in steep terrain (e.g., Rengers et al. 2020) or infrastructure risk assessment (e.g., Jong-Levinger

et al., 2024) require rapid information. In such cases, approaches that combine multispectral and SAR data, or utilize high-frequency, coarse-resolution sensors may be valuable (e.g., BRISK, <https://cimss.ssec.wisc.edu/brisk/>; Orland et al. 2025).

Limitations and future opportunities

We validated burn-severity products using three complementary field metrics from forested ecosystems: CBI, BA loss, and tree mortality (Table 2). Consequently, our results may not extend to grasslands and other ecosystems with differing relationships between burn severity and satellite-derived indices (Stambaugh et al. 2015). Additionally, our field data do not include direct measures of soil effects, such as changes in water repellency, organic matter loss, or ash cover, despite their important influence on a range of ecosystem processes (Lewis et al. 2021; Busby et al. 2023; Baltzer et al. 2025; Fortin et al. 2026). However, CBI incorporates qualitative assessments of soil burn severity, which are difficult to capture remotely; this may partially explain why CBI models were less accurate than models focused on BA loss or tree mortality (Fig. 5). Field measurements of burn severity were also collected 0–3 years after fire. Although we accounted for collection timing statistically, delayed mortality and early recovery introduce ecological variability that can be difficult to capture with post-fire imagery alone (Reilly et al. 2023). Finally, our analyses focused on moderate-resolution multispectral imagery and C-band SAR because they are freely available and are widely used for burn-severity mapping. Emerging missions such as NISAR (NASA-ISRO Synthetic Aperture Radar) may further improve observational latency and mapping accuracy, particularly for detecting soil effects and changes in vegetation structure.

Conclusion

This study provides new insight into the atmospheric and surface conditions that delay post-fire assessments in Canada and the United States, as well as differences in observational latency, coverage, and accuracy among commonly used remotely sensed datasets. These findings can help agency staff and practitioners select datasets that provide timely, reliable information to support post-fire decision-making. We found that meteorological and atmospheric conditions create distinct regional challenges, with cloud cover emerging as the most persistent source of image obstruction, particularly in mesic and coastal regions where it substantially limits multispectral data coverage. Our results also show that Sentinel-1 provides rapid, unobstructed post-fire coverage, whereas burn-severity estimates derived from HLS, Landsat, and Sentinel-2 correspond more closely with field

observations. Consequently, although it is not yet widely used for burn-severity mapping, HLS offers the strongest combination of timeliness and accuracy among the datasets evaluated here and represents a promising option for operational post-fire assessment. As fire regimes continue to shift under warmer and drier conditions, the utility of remote-sensing datasets will continue to evolve. Nevertheless, freely available moderate-resolution satellite imagery will remain essential for assessing fire effects, evaluating post-fire risk, and prioritizing management actions.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s42408-026-00589-y>.

Supplementary Material 1.

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Authors' contributions

K.C.R. conceived the ideas, performed analyses, and wrote the initial draft. E.W., A.A.H., A.S., K.J.B., C.A.B., Y.B., S.U.B., J.T.C., K.D.C., M.J.C., J.E.C., V.D., P.R.D., K.T.D., B.J.H., Z.A.H., S.M.H., J.M.I., P.M.A.J., M.R.K., M.M.L., K.R.M., J.E.M., M.-A.P., J.P.R., S.J.S., C.S.S., M.T.S., N.T., E.M.N.T., S.T.V., A.G.V., K.W., B.D.W., J.W., J.T.W., E.L.Y., & L.L.Y. provided input into methods, contributed data, and reviewed the draft.

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Data availability

Novel datasets and analytical code supporting this article are available in the Zenodo repository at: <https://doi.org/10.5281/zenodo.19741104>.

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